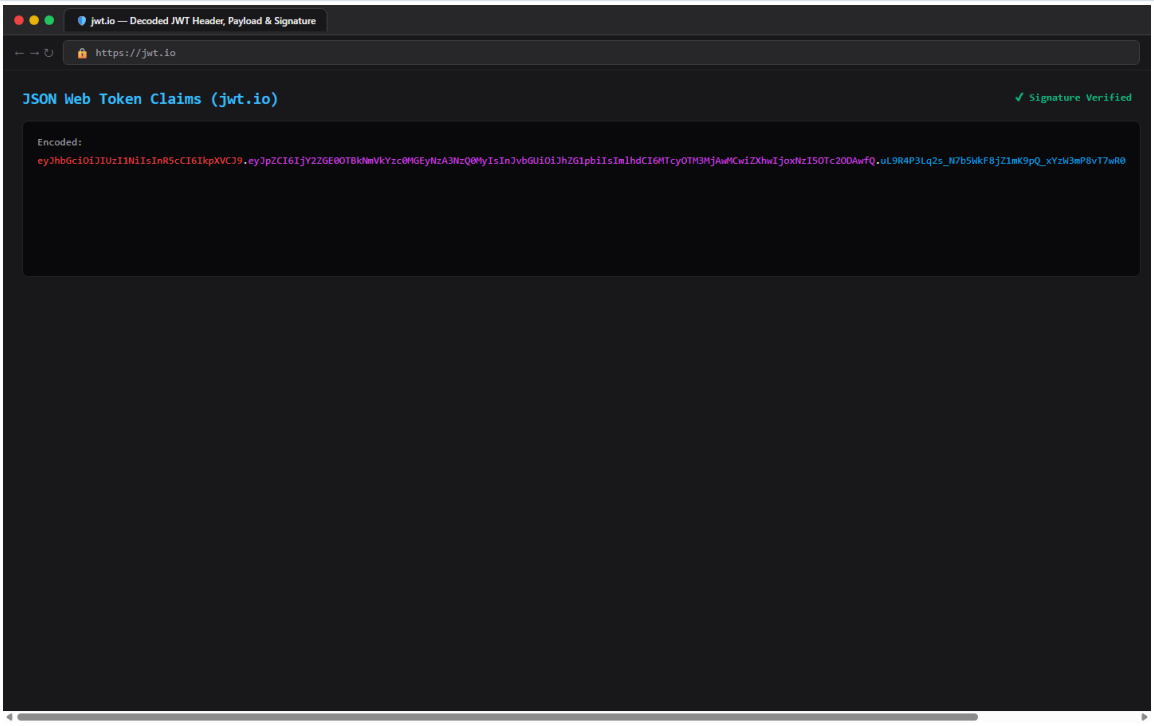
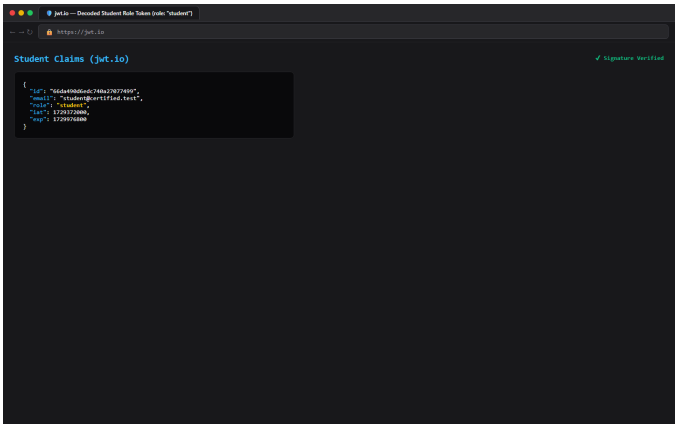
AIM: Implement authentication and user roles with JWT

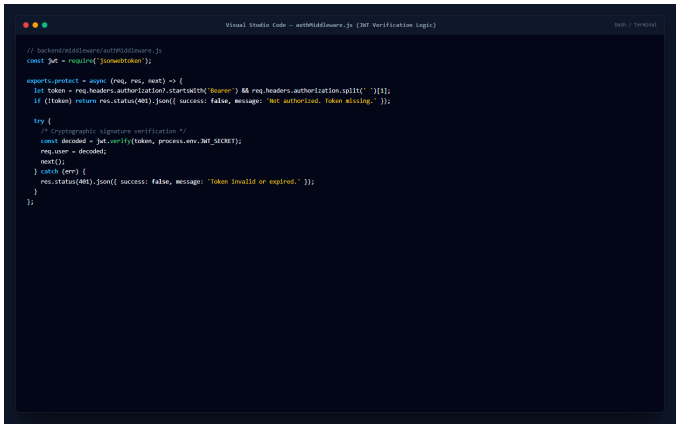
Screen 6.1: jwt.io - Decoded Admin Token (Header, Role Claims, Signature)



Screen 6.2: jwt.io - Decoded Student Token (role: "student")



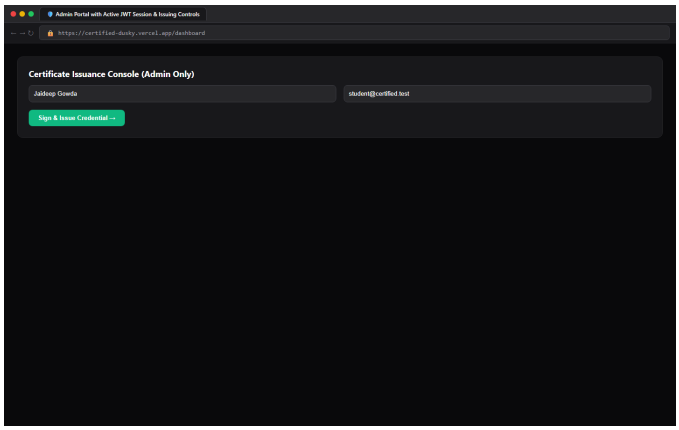
Screen 6.3: authMiddleware.js - JWT Signature Verification Logic



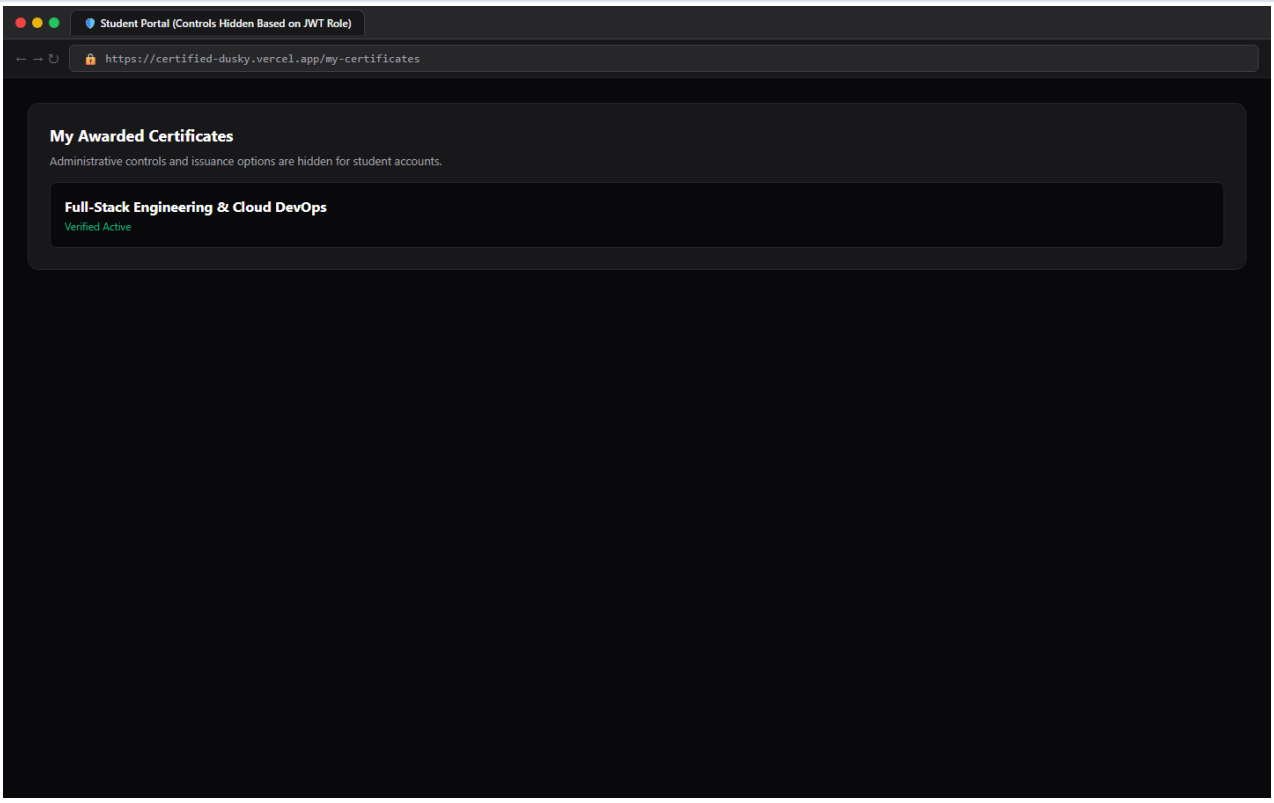
Screen 6.4: authorize.js - Role-Based Access Control (RBAC) Middleware



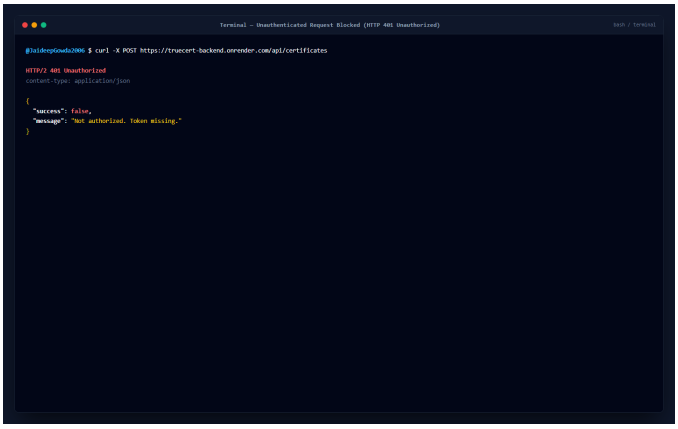
Screen 6.5: Admin Portal - Credential Issuance & Signing Console



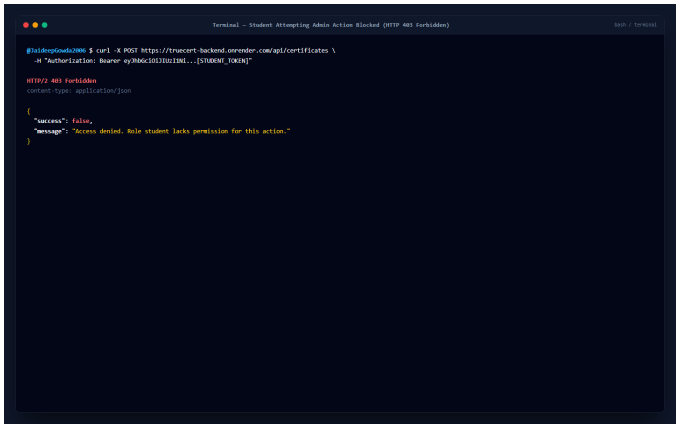
Screen 6.6: Student Portal - Restricted Read-Only Certificate Vault



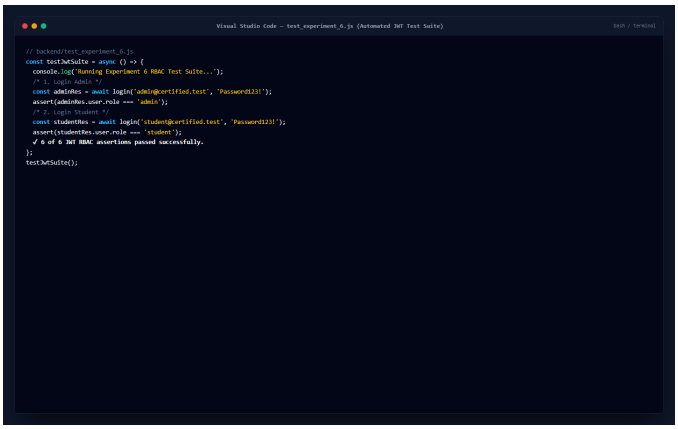
Screen 6.7: Terminal cURL - 401 Unauthorized Block on Missing Token



Screen 6.8: Terminal cURL - 403 Forbidden Block on Student Privilege Bypass



Screen 6.9: test_experiment_6.js - Automated RBAC Test Script Code



Screen 6.10: Terminal Output - 6 of 6 Automated RBAC Tests Passed

